



INTERNATIONAL
GRAMMAR SCHOOL
UNITY THROUGH DIVERSITY

2020 PRELIMINARY EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 1.5 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In questions 11 – 13, show relevant mathematical reasoning and / or calculations

Total Marks: 60 **Section I – 10 marks** (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 45 marks (pages 6–17)

- Attempt Questions 11–13
- Allow about 1 hour and 15 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 A PIN is to be chosen using four of the digits 0 through 9. Digits cannot be repeated.

How many possible PINs are there?

- A. 256
- B. 210
- C. 10 000
- D. 5 040

- 2 The function $f(x)$ is defined as $f(x) = x^3 + 2$

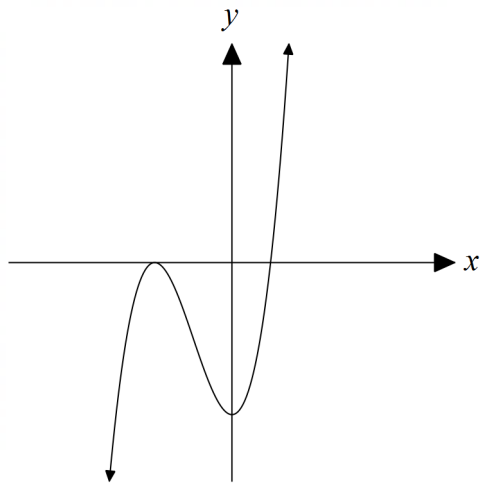
Which expression is equal to $f^{-1}(x)$?

- A. $\frac{1}{x^3} - 2$
- B. $\frac{1}{x^3 - 2}$
- C. $\sqrt[3]{x} - 2$
- D. $\sqrt[3]{x - 2}$

- 3 What is the domain of $2\cos^{-1}(3x)$?

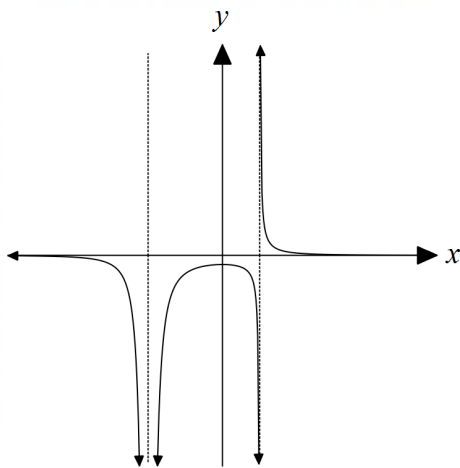
- A. $[-3, 3]$
- B. $[0, \pi]$
- C. $\left[-\frac{1}{3}, \frac{1}{3}\right]$
- D. $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

- 4 The graph of the function $f(x)$ is shown below.

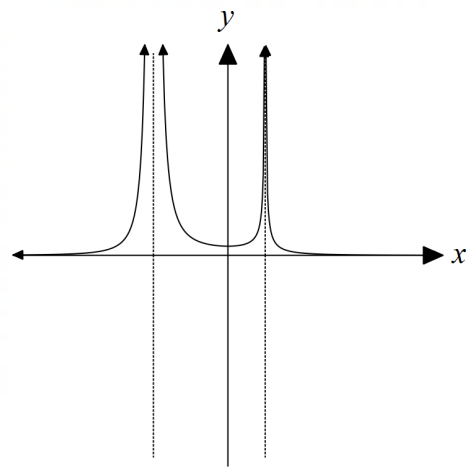


Which of the following graphs best represents $\frac{1}{f(x)}$?

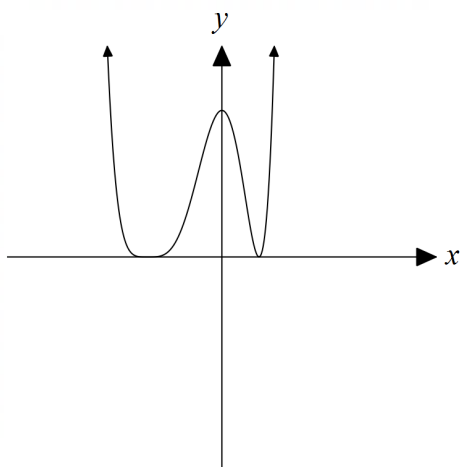
A.



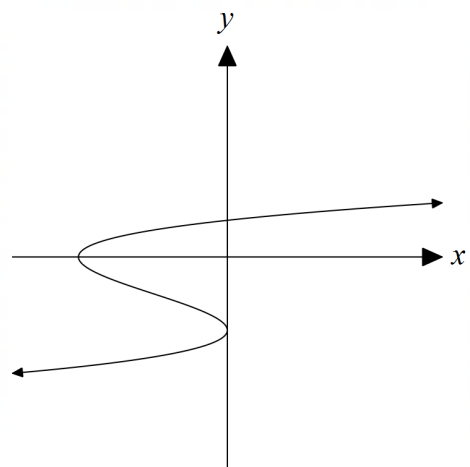
B.



C.

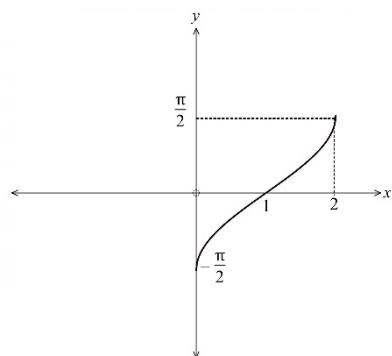


D.

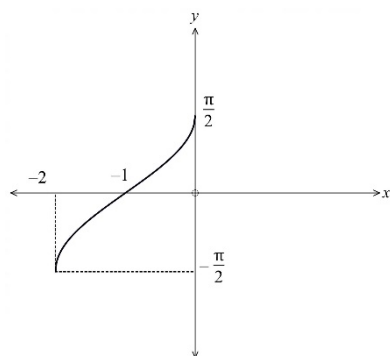


5 Which graph shows the curve $y = \sin^{-1}(x+1)$?

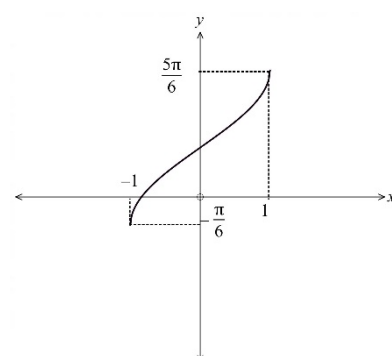
A.



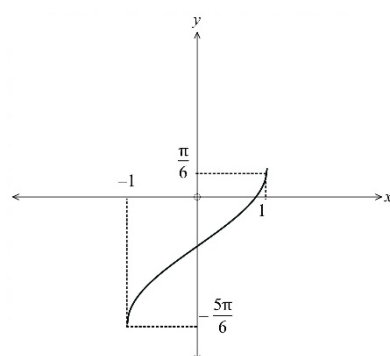
B.



C.



D.



6 Which of the following is a factor of $6x^3 - 5x^2 - 34x - 15$?

A. $x+1$

B. $x-3$

C. $x-2$

D. $x+5$

7 Given $N = 50 + 200e^{-kt}$, where k is a positive constant, what value does N approach as t approaches infinity?

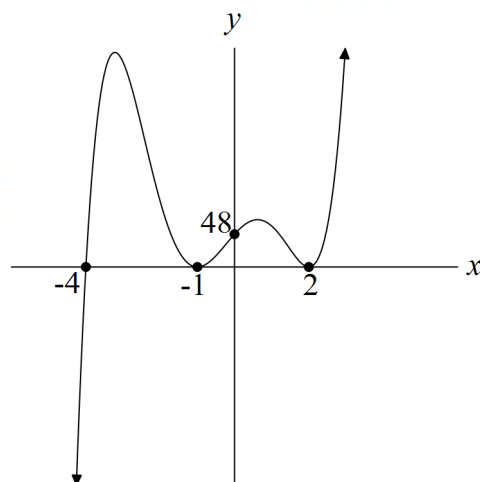
A. 50

B. 200

C. 250

D. 150

- 8 The diagram shows the graph of $y = a(x+b)^2(x+c)^2(x+d)$.



What are possible values of a , b , c and d ?

- A. $a = 3$, $b = 4$, $c = 1$, $d = 2$
 - B. $a = 6$, $b = 4$, $c = 1$, $d = 2$
 - C. $a = 3$, $b = -2$, $c = 1$, $d = 4$
 - D. $a = 6$, $b = -2$, $c = 1$, $d = 4$
- 9 What is the coefficient of x^7 in the expansion of $\left(x + \frac{4}{x}\right)^{13}$?

- A. 18 304
- B. 45 760
- C. 286
- D. 64

10 The gradient of the tangent to the curve $y = 2\cos\left(\frac{x}{2}\right)$ at the point

where $x = \frac{\pi}{3}$ is $-\frac{1}{2}$.

What is the gradient on the curve where $x = \frac{11\pi}{3}$?

Hint: sketch the function.

A. $-\frac{1}{2}$

B. $\frac{1}{2}$

C. -2

D. 2

Section II

60 marks

Attempt Questions 11–13

Allow about 1 hour and 15 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

In Questions 11–13, your responses should include relevant mathematical reasoning and/or calculations.

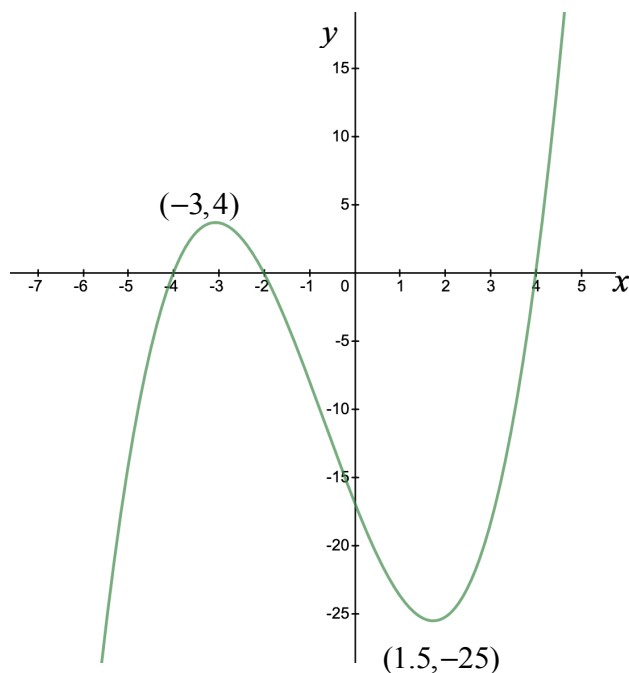
Question 11 (16 marks) Use the Question 11 Writing Booklet.

- (a) Solve $|3x - 6| \geq 15$ **2**
- (b) Write the expansion of $(x - 2y)^4$. **2**
- (c) What is the exact value of $\tan^{-1}\left(2\sin\left(-\frac{2\pi}{3}\right)\right)$? **2**
- (d) If $\sin\theta = \frac{5}{6}$ for $\frac{\pi}{2} \leq \theta \leq \pi$, what is the exact value of $\sin 2\theta$? **2**
- (e) Four Year 12 students, three Year 11 students, and two Year 10 students are seated around a circular table randomly.
- (i) How many seating arrangements are possible? **1**
- (ii) What is the probability that the three groups are seated together? **2**
- (f) The polynomial $P(x) = x^3 - 10x^2 - 39x + 180$ has roots α , β , γ .
- (i) Find $\alpha^2 + \beta^2 + \gamma^2$. **2**
- (ii) Find $(\alpha\beta)^{-2} + (\alpha\gamma)^{-2} + (\beta\gamma)^{-2}$. **3**

Question 12 (14 marks) Use the Question 12 Writing Booklet.

(a) Show that $\sin \theta \sin \left(\frac{\pi}{2} - \theta \right) = \frac{1}{2} \cos \left(2\theta - \frac{\pi}{2} \right)$ **2**

(b) The diagram shows the graph of a cubic function $y = f(x)$.



Draw separate one-third of a page sketches of the following functions, clearly showing any transformations of the turning points and x- and y-intercepts, as applicable.

(i) $y = f(|x|)$ **2**

(ii) $y^2 = f(x)$ **2**

Question 12 continues on page 9

Question 12 (continued)

- (c) A particle is moving along the x -axis. Its position, x metres, at time t seconds is given by $x(t) = -\frac{t^3}{3} + \frac{11t^2}{2} - 28t + 2$ for $t \geq 0$.

(i) When is the particle at rest? **2**

(ii) For what values of t is the particle moving to the right? **1**

- (d) A curve in the xy -plane is defined parametrically by $x = 4t^2$ and $y = 8t$, where t is the parameter.

(i) Draw the curve for $-3 \leq t \leq 3$, showing the beginning and ending coordinates and any intercepts. **2**

(ii) For what value or values of t will the parabola intersect the line $y = 6 - 2x$? **3**

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet.

- (a) Five points are scattered over a $2m$ by $2m$ square. Show that there is at least one pair of points which are a distance at most $\sqrt{2}m$ apart. **2**

- (b) Given $t = \tan 15^\circ$,

(i) show that $\frac{2t}{1-t^2} = \frac{1}{\sqrt{3}}$. **1**

- (ii) Hence, find the exact value of $\tan 15^\circ$. **2**

- (c) A bottle of soft drink has been taken out of a refrigerator and is initially 2°C . The soft drink is placed on a counter in a room, where the ambient temperature is 28°C .

After being in the room for 10 minutes, the temperature of the soft drink is 6°C .

The change in temperature of the soft drink can be modelled by

$\frac{dT}{dt} = k(T - 28)$, where T is the temperature of the soft drink, t is the time in minutes after the soft drink is taken out of the refrigerator, and k is a constant.

- (i) Show that $T = 28 + Ae^{kt}$, where A is a constant, satisfies $\frac{dT}{dt} = k(T - 28)$. **1**

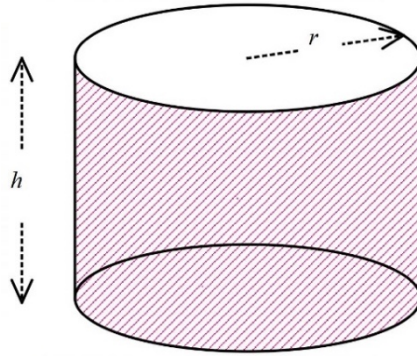
- (ii) Show that $k = -0.0167$ (3sf) **2**

- (iii) After 30 minutes, at what rate is the temperature of the soft drink changing? **2**

Question 13 continues on page 11

Question 13 (continued)

- (d) The solid shown below is a cylinder which has its height equal to its diameter.



A virtual 3-D model is created which maintains the ratio of height and diameter.

In an animation, the model is being scaled up so that its volume is increasing at a steady rate of 120 cm^3 per second.

Note: The volume of a cylinder is given by $V = \pi r^2 h$

- (i) Find the rate at which the radius is increasing when it reaches 2 cm. **4**

- (ii) What is the radius when $\frac{dr}{dt} = \frac{80}{\pi} \text{ cm s}^{-1}$? **1**

End of paper

Mathematics Extension

Section I Multiple-Choice Answer Sheet

1 A ☐ B ☐ C ☐ D ☐

2 A ☐ B ☐ C ☐ D ☐

3 A ☐ B ☐ C ☐ D ☐

4 A ☐ B ☐ C ☐ D ☐

5 A ☐ B ☐ C ☐ D ☐

6 A ☐ B ☐ C ☐ D ☐

7 A ☐ B ☐ C ☐ D ☐

8 A ☐ B ☐ C ☐ D ☐

9 A ☐ B ☐ C ☐ D ☐

10 A ☐ B ☐ C ☐ D ☐



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Use the multiple-choice answer sheet for Questions 1–10.

- 1 A PIN is to be chosen using four of the digits 0 through 9. Digits cannot be repeated.

How many possible PINs are there?

A. 210

B. 256

C. 10 000

☒ D. 5 040

$$10 \times 9 \times 8 \times 7 = 5040$$

Or

$${}^{10}P_4 = 5040$$

- 2 The function $f(x)$ is defined as $f(x) = x^3 + 2$

Which expression is equal to $f^{-1}(x)$?

A. $\frac{1}{x^3} - 2$

B. $\frac{1}{x^3 - 2}$

C. $\sqrt[3]{x} - 2$

☒ D. $\sqrt[3]{x - 2}$

$$y^3 + 2 = x$$

$$y^3 = x - 2$$

$$y = \sqrt[3]{x - 2}$$

- 3 What is the domain of $2\cos^{-1}(3x)$?

A. $[-3, 3]$

B. $[0, \pi]$

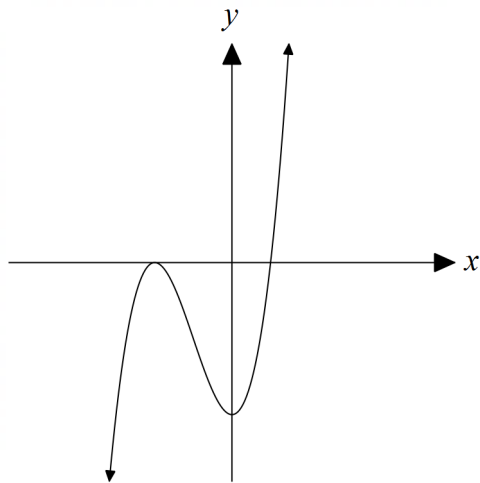
☒ C. $\left[-\frac{1}{3}, \frac{1}{3}\right]$

D. $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$$-1 \leq 3x \leq 1$$

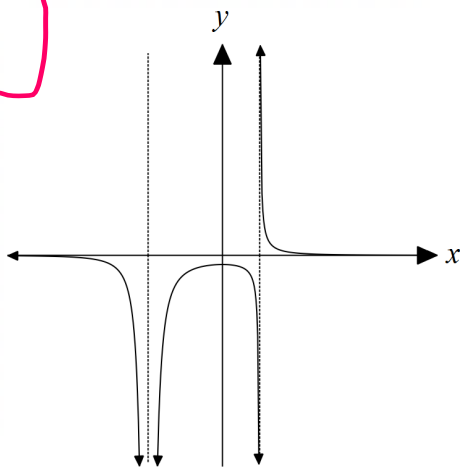
$$\frac{-1}{3} \leq x \leq \frac{1}{3}$$

- 4 The graph of the function $f(x)$ is shown below.

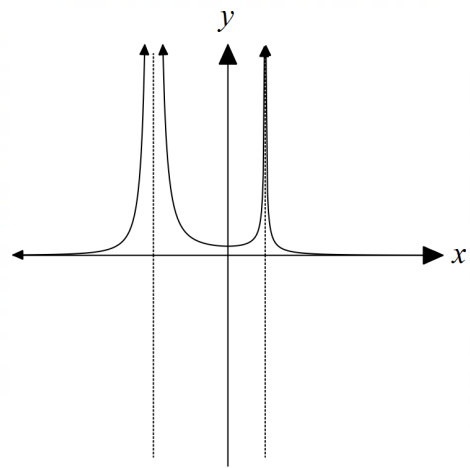


Which of the following graphs best represents $\frac{1}{f(x)}$?

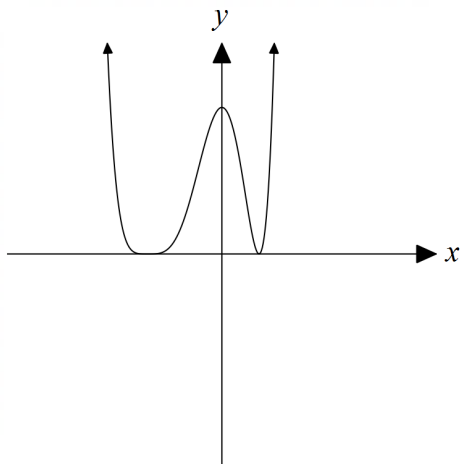
A.



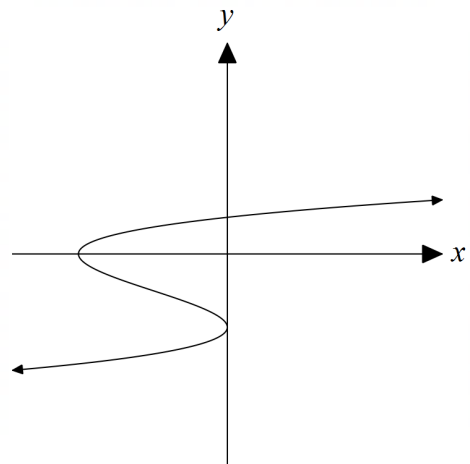
B.



C.

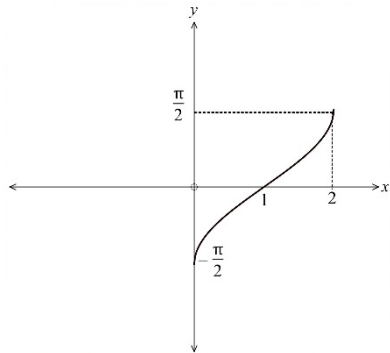


D.

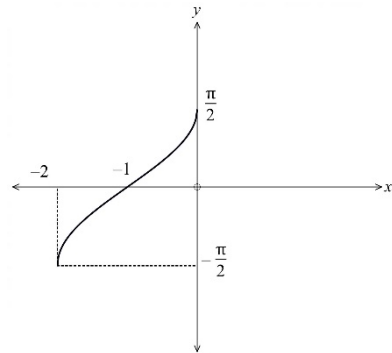


5 Which graph shows the curve $y = \sin^{-1}(x+1)$?

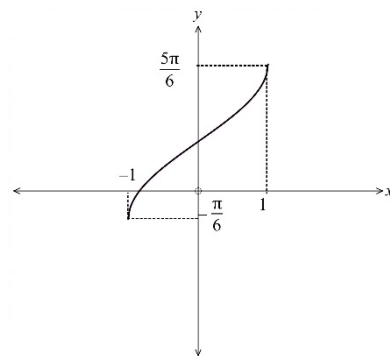
A.



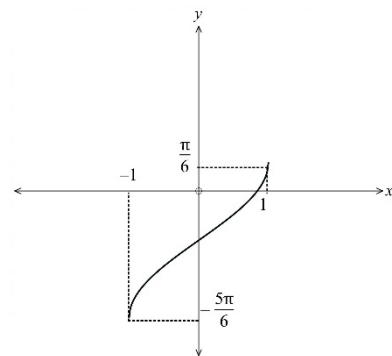
B.



C.



D.



Transformation is 1 to the left

6 Which of the following is a factor of $6x^3 - 5x^2 - 34x - 15$?

A. $x+1$

B. $x-3$

C. $x-2$

D. $x+5$

$$6 \times 3^3 - 5 \times 3^2 - 34 \times 3 - 15 = 0$$

7 Given $N = 50 + 200e^{-kt}$, where k is a positive constant, what value does N approach as t approaches infinity?

A. 50

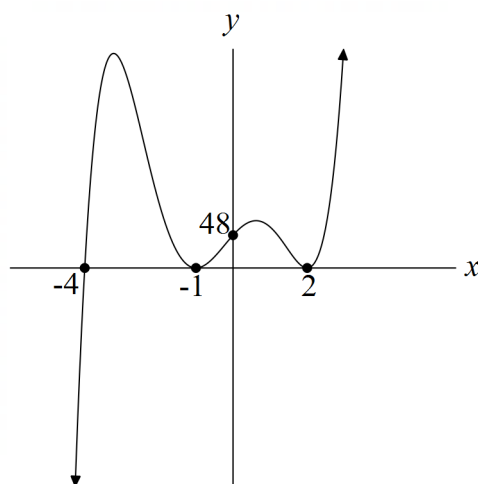
B. 200

C. 250

D. 150

$$\begin{aligned} \lim_{t \rightarrow \infty} (e^{-kt}) &= 0 \\ \lim_{t \rightarrow \infty} (50 + 200e^{-kt}) &= 50 + 200 \times 0 \\ &= 50 \end{aligned}$$

- 8 The diagram shows the graph of $y = a(x+b)^2(x+c)^2(x+d)$.



What are possible values of a , b , c and d ?

A. $a = 3, b = 4, c = 1, d = 2$

B. $a = 6, b = 4, c = 1, d = 2$

C. $a = 3, b = -2, c = 1, d = 4$

D. $a = 6, b = -2, c = 1, d = 4$

Single root at -4 , so $d=4$
Double root at -1 and 2 , so b and c are 1 and -2

$$48 = a(b)^2(c)^2(d)$$

$$48 = a \times (-2)^2(1)^2(4)$$

$$48 = a \times 16$$

$$a = 3$$

- 9 What is the coefficient of x^7 in the expansion of $\left(x + \frac{4}{x}\right)^{13}$?

A. 18 304

B. 45 760

C. 286

D. 64

$$\left(\begin{matrix} 13 \\ r \end{matrix}\right) x^{13-r} \left(\frac{4}{x}\right)^r$$

$$13 - r - r = 7$$

$$6 = 2r$$

$$r = 3$$

$$\left(\begin{matrix} 13 \\ 3 \end{matrix}\right) x^{10} \left(\frac{4}{x}\right)^3 = 18304x^7$$

10 The gradient of the tangent to the curve $y = 2\cos\left(\frac{x}{2}\right)$ at the point

where $x = \frac{\pi}{3}$ is $-\frac{1}{2}$.

What is the gradient on the curve where $x = \frac{11\pi}{3}$?

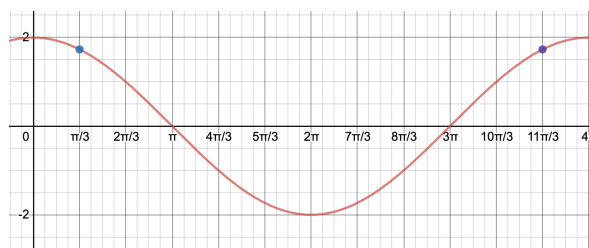
Hint: sketch the function.

A. $-\frac{1}{2}$

B. $\frac{1}{2}$

C. -2

D. 2



By the symmetry of the curve, it must be $\frac{1}{2}$

Question	Answer
1	D
2	D
3	C
4	A
5	B
6	B
7	A
8	C
9	A
10	B

Section II

60 marks

Attempt Questions 11–13

Allow about 1 hour and 15 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

In Questions 11–13, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (16 marks) Use the Question 11 Writing Booklet.

Question 11a	Marks
• Provides the correct solution	2
• Writes two inequalities based on the given inequality	1

(a) Solve $|3x - 6| \geq 15$ **2**

$$\begin{aligned}
 3x - 6 &\geq 15 & 3x - 6 &\leq -15 \\
 3x &\geq 21 & \text{or} & 3x &\leq -9 \\
 x &\geq 7 & & x &\leq -3
 \end{aligned}$$

Question 11b	Marks
• Provides the correct solution	2
• Uses the binomial theorem, or equivalent merit	1

(b) Write the expansion of $(x - 2y)^4$. **2**

$$\begin{aligned}
 &{}^4C_0x^4 + {}^4C_1x^3(-2y) + {}^4C_2x^2(-2y)^2 + {}^4C_3x(-2y)^3 + {}^4C_4(-2y)^4 \\
 &= x^4 - 8x^3y + 24x^2y^2 - 32xy^3 + 16y^4
 \end{aligned}$$

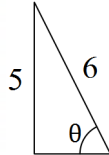
Question 11c	Marks
• Provides the correct solution	2
• Finds an angle that is a multiple of $\frac{\pi}{3}$.	1

(c) What is the exact value of $\tan^{-1}\left(2\sin\left(-\frac{2\pi}{3}\right)\right)$? **2**

$$\begin{aligned}
 \tan^{-1}\left(2\sin\left(-\frac{2\pi}{3}\right)\right) &= \tan^{-1}\left(2 \times \frac{-\sqrt{3}}{2}\right) \\
 &= \tan^{-1}(-\sqrt{3}) \\
 &= -\frac{\pi}{3}
 \end{aligned}$$

Question 11d	Marks
• Provides the correct solution	2
• Uses Pythagoras' Theorem to find the third side of the right triangle, or equivalent merit	1

- (d) If $\sin \theta = \frac{5}{6}$ for $\frac{\pi}{2} \leq \theta \leq \pi$, what is the exact value of $\sin 2\theta$? **2**



$$\begin{aligned}\sqrt{6^2 - 5^2} &= \sqrt{11} \\ \sin \theta &= \frac{5}{6}, \cos \theta = -\frac{\sqrt{11}}{6}, \\ \sin 2\theta &= 2 \sin \theta \cos \theta \\ &= 2 \times \frac{5}{6} \times -\frac{\sqrt{11}}{6} \\ &= -\frac{5\sqrt{11}}{18}\end{aligned}$$

- (e) Four Year 12 students, three Year 11 students, and two Year 10 students are seated around a circular table randomly.

Question 11e (i)	Marks
• Provides the correct solution	1

- (i) How many seating arrangements are possible? **1**

$$8! = 40320$$

Question 11e (ii)	Marks
• Provides the correct solution	2
• Finds the number of ways each group can be arranged, or equivalent merit	1

- (ii) What is the probability that the three groups are seated together? **2**

$$\begin{aligned}\frac{4! \times 3! \times 2! \times 2!}{40320} &= \frac{576}{40320} \\ &= \frac{1}{70}\end{aligned}$$

Question 11f (i)	Marks
• Provides the correct solution	2
• Uses $(\alpha + \beta + \gamma)^2$ or equivalent merit	1

(f) The polynomial $P(x) = x^3 - 10x^2 - 39x + 180$ has roots α , β , γ .

(i) Find $\alpha^2 + \beta^2 + \gamma^2$. **2**

$$(\alpha + \beta + \gamma)^2 = \alpha^2 + \alpha\beta + \alpha\gamma + \alpha\beta + \beta^2 + \gamma\beta + \alpha\gamma + \beta\gamma + \gamma^2$$

$$(\alpha + \beta + \gamma)^2 = (\alpha^2 + \beta^2 + \gamma^2) + 2(\alpha\beta + \alpha\gamma + \gamma\beta)$$

$$\left(\frac{-b}{a}\right)^2 = (\alpha^2 + \beta^2 + \gamma^2) + 2\left(\frac{c}{a}\right)$$

$$(10)^2 = (\alpha^2 + \beta^2 + \gamma^2) + 2 \times -39$$

$$\alpha^2 + \beta^2 + \gamma^2 = 178$$

Question 11f (ii)	Marks
• Provides the correct solution	3
• Makes significant progress towards the solution	2
• Expresses each term as a fraction and attempts to use a common denominator	1

(ii) Find $(\alpha\beta)^{-2} + (\alpha\gamma)^{-2} + (\beta\gamma)^{-2}$. **3**

$$\frac{1}{(\alpha\beta)^2} + \frac{1}{(\alpha\gamma)^2} + \frac{1}{(\beta\gamma)^2} = \frac{\gamma^2}{(\alpha\beta\gamma)^2} + \frac{\beta^2}{(\alpha\beta\gamma)^2} + \frac{\alpha^2}{(\alpha\beta\gamma)^2}$$

$$= \frac{\alpha^2 + \beta^2 + \gamma^2}{(\alpha\beta\gamma)^2}$$

$$= 178 \div \left(\frac{-d}{a}\right)^2$$

$$= 178 \times \frac{1}{32400}$$

$$= \frac{89}{16200}$$

Question 12 (14 marks) Use the Question 12 Writing Booklet.

Question 12a	Marks
• Provides the correct proof	2
• Applies an appropriate trigonometric identity, or equivalent merit	1

(a) Show that $\sin \theta \sin \left(\frac{\pi}{2} - \theta \right) = \frac{1}{2} \cos \left(2\theta - \frac{\pi}{2} \right)$ **2**

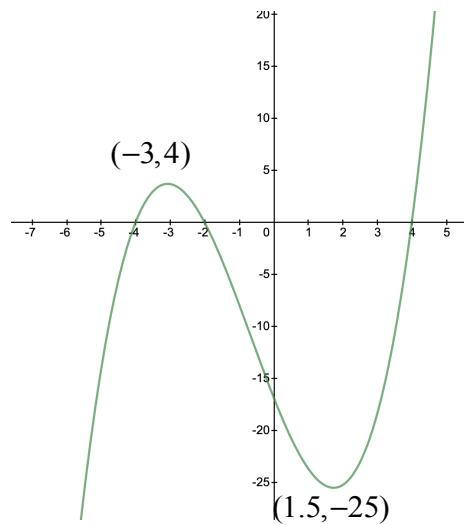
Method 1

$$\begin{aligned}
 LHS &= \sin \theta \sin \left(\frac{\pi}{2} - \theta \right) \\
 &= \frac{1}{2} \times 2 \sin \theta \cos \theta \\
 &= \frac{1}{2} \sin 2\theta \\
 &= \frac{1}{2} \cos \left(\frac{\pi}{2} - 2\theta \right) \\
 &= \frac{1}{2} \cos \left(2\theta - \frac{\pi}{2} \right) \\
 &= RHS
 \end{aligned}$$

Method 2

$$\begin{aligned}
 LHS &= \sin \theta \sin \left(\frac{\pi}{2} - \theta \right) \\
 &= \frac{1}{2} \left(\cos \left(\theta - \left(\frac{\pi}{2} - \theta \right) \right) - \cos \left(\theta + \left(\frac{\pi}{2} - \theta \right) \right) \right) \\
 &= \frac{1}{2} \left(\cos \left(2\theta - \frac{\pi}{2} \right) - \cos \left(\frac{\pi}{2} \right) \right) \\
 &= \frac{1}{2} \left(\cos \left(2\theta - \frac{\pi}{2} \right) - 0 \right) \\
 &= \frac{1}{2} \cos \left(2\theta - \frac{\pi}{2} \right) \\
 &= RHS
 \end{aligned}$$

(b) The cubic function $y = f(x)$ is graphed below.

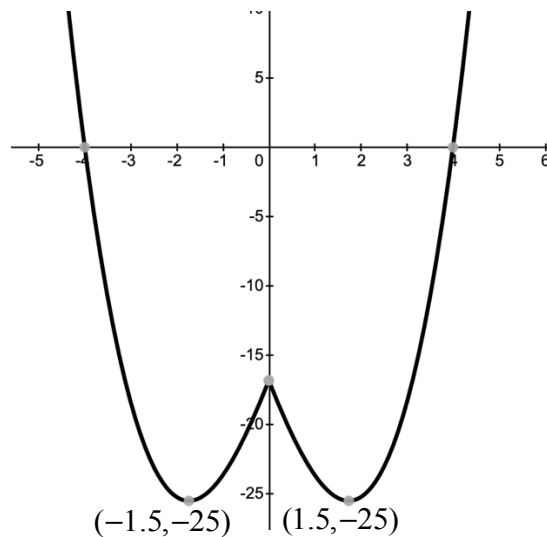


Question 12b (i)		Marks
• Provides the correct graph		2
• Graphs an absolute value graph based on the given graph		1

Draw separate one-third of a page sketches of the following functions, clearly showing any transformations of the turning points and x- and y-intercepts, as applicable.

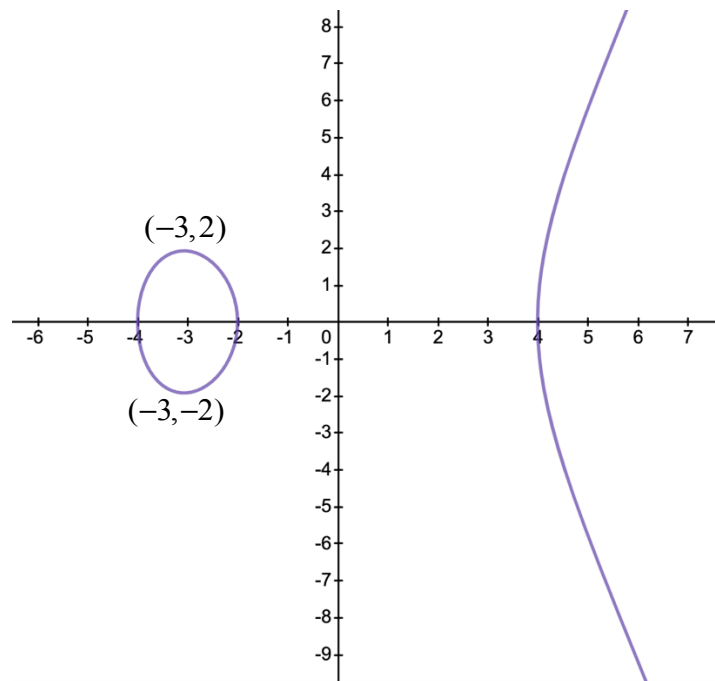
(i) $y = f(|x|)$

2



Question 12b (ii)	Marks
• Provides the correct graph	2
• Some sections of the graph correct	1

(ii) $y^2 = f(x)$ **2**



(c) A particle is moving along the x -axis. Its position, x metres, at time t seconds is given by $x(t) = -\frac{t^3}{3} + \frac{11t^2}{2} - 28t + 2$ for $t \geq 0$.

Question 12c (i)	Marks
• Provides the correct solution	2
• Attempts to solve the first derivative equal to 0	1

(i) When is the particle at rest? **2**

$$v(t) = -t^2 + 11t - 28$$

$$t^2 - 11t + 28 = 0$$

$$(t - 4)(t - 7) = 0$$

$$t = 4\text{ s}, t = 7\text{ s}$$

Question 12c (ii)	Marks
• Provides the correct solution	1

(ii) For what values of t is the particle moving to the right? **1**

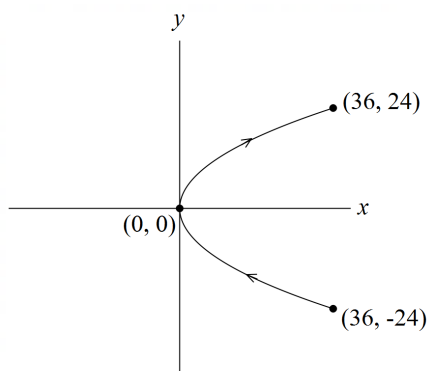
$$v(t) \geq 0$$

$$4\text{ s} < t < 7\text{ s}$$

- (d) A curve in the xy -plane is defined parametrically by $x = 4t^2$ and $y = 8t$, where t is the parameter.

Question 12d (i)	Marks
• Provides the correct graph, with end points and turning point labelled	2
• Provides the correct shape of the graph or x and y reversed	1

- (i) Draw the curve for $-3 \leq t \leq 3$, showing the beginning and ending coordinates and any intercepts. **2**



Question 12d (ii)	Marks
• Provides the correct solutions	3
• Provides the correct solutions in terms of x and y rather than t	2
• Develops a quadratic equation to find t	2
• Substitutes the parametric equations into the line, or equivalent merit	1

- (ii) For what value or values of t will the parabola intersect the line $y = 6 - 2x$? **3**

$$\begin{aligned}
 t &= \frac{y}{8} \\
 x &= 4 \times \left(\frac{y}{8} \right)^2 \\
 y^2 &= 16x \\
 t &= \frac{y}{8} \\
 8t &= 6 - 8t^2 \\
 8t^2 + 8t - 6 &= 0 \\
 2(4t^2 + 4t - 3) &= 0 \\
 2(4t^2 + 6t - 2t - 3) &= 0 \\
 2(2t(2t + 3) - (2t + 3)) &= 0 \\
 2(2t + 3)(2t - 1) &= 0 \\
 t &= \frac{-3}{2}, t = \frac{1}{2}
 \end{aligned}$$

Question 13 (15 marks) Use the Question 13 Writing Booklet.

Question 13a	Marks
• Provides the correct proof	2
• Correctly applies the Pigeonhole Principle, or equivalent merit	1

- (a) Five points are scattered over a $2m$ by $2m$ square. Show that there is at least one pair of points which are a distance at most $\sqrt{2}m$ apart. **2**

Divide the square into four $1m \times 1m$ squares.

By the pigeon hole principle, at least one of the squares must contain 2 points. The furthest distance between the 2 points is

the diagonal of the smaller square, which is $\sqrt{2}m$

- (b) Given $t = \tan 15^\circ$,

Question 13b (i)	Marks
• Provides a correct proof	1

- (i) show that $\frac{2t}{1-t^2} = \frac{1}{\sqrt{3}}$. **1**

$$\tan 30 = \frac{2 \tan 15}{1 - \tan^2 15}$$

$$\frac{1}{\sqrt{3}} = \frac{2t}{1-t^2}$$

Question 13b (ii)	Marks
• Provides the correct solution	2
• Develops and attempts to solve a quadratic equation, or equivalent merit	1

- (ii) Hence, find the exact value of $\tan 15^\circ$. **2**

$$\begin{aligned} \frac{2t}{1-t^2} &= \frac{1}{\sqrt{3}} \\ 2\sqrt{3}t &= 1-t^2 \\ t^2 + 2\sqrt{3}t - 1 &= 0 \end{aligned} \quad \begin{aligned} t &= \frac{-2\sqrt{3} \pm \sqrt{(2\sqrt{3})^2 + 4}}{2} \\ &= -\sqrt{3} \pm 2 \end{aligned}$$

- (c) A bottle of soft drink has been taken out of a refrigerator and is initially 2°C . The soft drink is placed on a counter in a room, where the ambient temperature is 28°C .

After being in the room for 10 minutes, the temperature of the soft drink is 6°C .

The change in temperature of the soft drink can be modelled by

$\frac{dT}{dt} = k(T - 28)$, where T is the temperature of the soft drink, t is the time in minutes after the soft drink is taken out of the refrigerator, and k is a constant.

Question 13c (i)	Marks
• Provides correct proof	1

- (i) Show that $T = 28 + Ae^{kt}$, where A is a constant, satisfies

$$\frac{dT}{dt} = k(T - 28).$$

1

$$T = 28 + Ae^{kt}$$

$$\frac{dT}{dt} = kAe^{kt}$$

$$\frac{dT}{dt} = k(T - 28)$$

Question 13c (ii)	Marks
• Provides the correct solution	2
• Finds A , or equivalent merit.	1

- (ii) Show that $k = -0.0167$ (3sf)

2

$$t = 0, T = 2$$

$$T = 28 + Ae^{kt}$$

$$2 = 28 + A$$

$$A = -26$$

$$t = 10, T = 6$$

$$6 = 28 - 26e^{k \times 10}$$

$$-22 = -26e^{10k}$$

$$e^{10k} = \frac{11}{13}$$

$$10k = \ln \frac{11}{13}$$

$$k = \frac{1}{10} \ln \frac{11}{13}$$

$$= -0.0167 \text{ (3sf)}$$

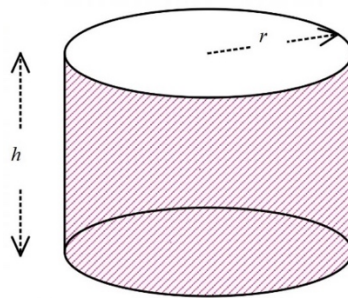
Question 13c (iii)		Marks
• Provides the correct solution		2
• Finds the temperature at $t = 30$, or equivalent merit		1

- (iii) After 30 minutes, at what rate is the temperature of the soft drink changing?

2

$$\begin{aligned}
 T &= 28 - 26e^{-0.0167 \times 30} \\
 &= 12.25 \\
 \frac{dT}{dt} &= -0.0167(12.25 - 28) \\
 &= 0.263^\circ\text{C} / \text{min}
 \end{aligned}
 \quad \text{or} \quad
 \begin{aligned}
 \frac{dT}{dt} &= kAe^{kt} \\
 &= -0.0167 \times (-26)e^{-0.0167 \times 30} \\
 &= 0.263^\circ\text{C} / \text{min}
 \end{aligned}$$

- (d) The solid shown below is a cylinder which has its height equal to its diameter.



A virtual 3-D model is created which maintains the ratio of height and diameter.

In an animation, the model is being scaled up so that its volume is increasing at a steady rate of 120 cm^3 per second.

Note: The volume of a cylinder is given by $V = \pi r^2 h$

Question 13d (i)		Marks
• Provides the correct solution with units		4
• Correctly uses $\frac{dV}{dt} = 120$		3
• Differentiates with respect to t or r correctly, or equivalent merit		2
• Substitutes $2r = h$ into volume formula		1

- (i) Find the rate at which the radius is increasing when it reaches 2 cm.

4

$$\begin{aligned}
 V &= 2\pi r^2(2r) \\
 &= 4\pi r^3 \\
 \frac{dV}{dt} &= 6\pi r^2 \frac{dr}{dt} \\
 120 &= 6\pi(2)^2 \frac{dr}{dt} \\
 \frac{dr}{dt} &= \frac{5}{\pi} \text{ cm s}^{-1}
 \end{aligned}
 \quad \text{or} \quad
 \begin{aligned}
 \frac{dV}{dr} &= 6\pi r^2 \\
 \frac{dV}{dr} \times \frac{dr}{dt} &= \frac{dV}{dt} \\
 6\pi(2)^2 \times \frac{dr}{dt} &= 120 \\
 \frac{dr}{dt} &= \frac{5}{\pi} \text{ cm s}^{-1}
 \end{aligned}$$

Question 13d (ii)	Marks
• Provides the correct solution	1

(ii) What is the radius when $\frac{dr}{dt} = \frac{80}{\pi} \text{ cm s}^{-1}$? **1**

$$\begin{aligned}
 V &= 2\pi r^3 \\
 \frac{dV}{dt} &= 6\pi r^2 \frac{dr}{dt} & \text{or} & \quad V = 2\pi r^3 \\
 120 &= 6\pi r^2 \frac{80}{\pi} & \frac{dV}{dr} &= 6\pi r^2 \\
 120 &= 6r^2 \times 80 & \frac{dV}{dr} &= \frac{dV}{dr} \times \frac{dr}{dt} \\
 & & 120 &= 6\pi r^2 \times \frac{80}{\pi} \\
 & & 120 &= 6r^2 \times 80
 \end{aligned}$$

$$r^2 = \frac{1}{4}$$

$$r = \frac{1}{2} \text{ cm}$$

(can't be negative)

End of paper

Mapping Grid

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
1	1	ME-A1 Working with Combinatorics	ME11-5	E2
2	1	ME-F1 Further Work with Functions	ME11-2	E2
3	1	ME-T1 Inverse Trigonometric Functions	ME11-3	E2
4	1	ME-F1 Further Work with Functions	ME11-2	E3
5	1	ME-T1 Inverse Trigonometric Functions	ME11-3	E2
6	1	ME-F2 Polynomials	ME11-2	E3
7	1	ME-C1 Rates of Change	ME11-4	E3
8	1	ME-F2 Polynomials	ME11-2	E3
9	1	ME-A1 Working with Combinatorics	ME11-5	E3
10	1	MA-C1 Introduction to Differentiation MA-T1 Trigonometry and Measure of Angles	ME11-1	E4
11a	2	ME-F1 Further Work with Functions	ME11-2	E2
11b	2	ME-A1 Working with Combinatorics	ME11-5	E2
11c	2	ME-T1 Inverse Trigonometric Functions	ME11-3	E2-E3
11d	2	ME-T2 Further Trigonometric Identities	ME11-3	E2-E3
11e (i)	1	ME-A1 Working with Combinatorics	ME11-5	E2
11e (ii)	2	ME-A1 Working with Combinatorics MA-S1 Probability and Discrete Probability Distributions	ME11-5	E2-E3
11f (i)	2	ME-F2 Polynomials	ME11-2	E3
11f (ii)	3	ME-F2 Polynomials	ME11-2	E3
12a	2	ME-T2 Further Trigonometric Identities	ME11-3	E2-E3
12b (i)	2	ME-F1 Further Work with Functions	ME11-1 ME11-2	E2-E3
12b (ii)	2	ME-F1 Further Work with Functions	ME11-1 ME11-2	E2-E3
12c (i)	2	ME-C1 Rates of Change	ME11-4	E2
12c (ii)	1	ME-C1 Rates of Change ME-F1 Further Work with Functions	ME11-4 ME11-7	E2-E3
12d (i)	2	ME-F1 Further Work with Functions	ME11-2	E2-E3
12d (ii)	3	ME-F1 Further Work with Functions	ME11-2	E3
13a	2	ME-A1 Working with Combinatorics	ME11-1 ME11-5 ME11-7	E3-E4
13b (i)	1	ME-T2 Further Trigonometric Identities	ME11-3	E2
13b (ii)	2	ME-T2 Further Trigonometric Identities	ME11-1 ME11-3	E2-E3
13c (i)	1	ME-C1 Rates of Change	ME11-4	E2
13c (ii)	2	ME-C1 Rates of Change	ME11-4	E3
13c (iii)	2	ME-C1 Rates of Change	ME11-4	E3
13d (i)	4	ME-C1 Rates of Change	ME11-1 ME11-4	E2-E4
13d (ii)	1	ME-C1 Rates of Change	ME11-4	E2

Mathematics Extension

Section I Multiple-Choice Answer Sheet

- | | | | | |
|----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☒ |
| 2 | A ☐ | B ☐ | C ☐ | D ☒ |
| 3 | A ☐ | B ☐ | C ☒ | D ☐ |
| 4 | A ☒ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☒ | C ☐ | D ☐ |
| 6 | A ☐ | B ☒ | C ☐ | D ☐ |
| 7 | A ☒ | B ☐ | C ☐ | D ☐ |
| 8 | A ☐ | B ☐ | C ☒ | D ☐ |
| 9 | A ☒ | B ☐ | C ☐ | D ☐ |
| 10 | A ☐ | B ☒ | C ☐ | D ☐ |